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UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2018 INVITATIONAL B

Biology		Chemistry		Physics	
B01.	B	C01.	B	P01.	A
B02.	A	C02.	E	P02.	B
B03.	B	C03.	D	P03.	E
B04.	E	C04.	D	P04.	A
B05.	C	C05.	C	P05.	C
B06.	E	C06.	C	P06.	C
B07.	B	C07.	D	P07.	D
B08.	E	C08.	A	P08.	A
B09.	A	C09.	D	P09.	C
B10.	D	C10.	A	P10.	D
B11.	C	C11.	B	P11.	B
B12.	D	C12.	C	P12.	E
B13.	C	C13.	A	P13.	A
B14.	A	C14.	C	P14.	C
B15.	E	C15.	D	P15.	A
B16.	D	C16.	C	P16.	B
B17.	D	C17.	E	P17.	D
B18.	C	C18.	C	P18.	E
B19.	B	C19.	B	P19.	E
B20.	A	C20.	A	P20.	C

CHEMISTRY SOLUTIONS – UIL INVITATIONAL B 2018

C01. (B) An atom that is sp^3 hybridized always has four areas of electron density around it (either bonded atoms or non-bonding electron pairs).

C02. (E) $CuSO_4{\cdot}5H_2O$ has 9 moles of oxygen atoms in every mole of compound. Therefore 5 moles of compound will contain 5 × 9 = 45 moles of oxygen atoms.

C03. (D) A spontaneous reaction is one that is thermodynamically favorable (has a negative ΔG), meaning there is a net energy loss.

C04. (D) Assume you have 100.0 g of the substance. Then you have 43.7 g of P and 56.3 g of O. Dividing the mass of each element by its molar mass yields 1.411 mol of P and 3.519 mol of O. Divide both mole values by the smaller of the two values to convert at least one of them to an integer – this yields a ratio of 1 P atom for every 2.494 O atoms. Double both of these values to convert them both to integers and the result is 2 P atoms for every 5 O atoms, or P_2O_5. Answer (B), P_4O_{10}, has the same mole ratio, but an empirical formula is always the lowest whole-number ratio of atoms.

C05. (C) Since 6 CN^- react with each Fe^{3+}, if the reaction goes to completion it will consume 6×10^{-3} M CN^-, leaving 0.144 M CN^- unreacted in the solution.

C06. (C) The balanced equation for the reaction is

$$2\ B_5H_9 + 12\ O_2 \rightarrow 5\ B_2O_3 + 9\ H_2O$$

ΔH_{rxn} can be calculated from the heats of formation of the reactants and products:

$$\Delta H_{\text{rxn}} = \sum \Delta H_{\text{f}}\,(products) - \sum \Delta H_{\text{f}}\,(reactants)$$

$$\Delta H_{\text{rxn}} = [5 \times \Delta H_{\text{f}}\,(\text{B}_2\text{O}_3) + 9 \times \Delta H_{\text{f}}\,(\text{H}_2\text{O})] - [2 \times \Delta H_{\text{f}}\,(\text{B}_5\text{H}_9) + 12 \times \Delta H_{\text{f}}\,(\text{O}_2)]$$

The ΔH_f values for B_5H_9, B_2O_3, and H_2O (both liquid and gas) are given on the data page. Since ΔH_f refers to the compounds in their standard state (1 atm pressure and 25 °C), the value for $H_2O(l)$ should be used in the calculation, not the value for $H_2O(g)$. ΔH_f for O_2 is not given because ΔH_f for elements in their standard state is 0. Remember to multiply each heat of formation by the s stoichiometric coefficients for each species to get the total heat.

$$\Delta H_{\text{rxn}} = \left[5\text{ mol}\left(-1264\frac{\text{kJ}}{\text{mol}}\right) - 9\text{ mol}\left(-285.8\frac{\text{kJ}}{\text{mol}}\right)\right] - \left[2\text{ mol}\left(73.2\frac{\text{kJ}}{\text{mol}}\right) + 12\text{ mol}\left(0\frac{\text{kJ}}{\text{mol}}\right)\right]$$

$$\Delta H_{\text{rxn}} = -8892.2\text{ kJ} - 146.4\text{ kJ}$$

$$\Delta H_{\text{rxn}} = -9038.6\text{ kJ}$$

This is the ΔH_{rxn} for the balanced equation, but the standard enthalpy change of combustion is defined for one mole of the compound. The balanced reaction includes 2 moles of B_5H_9, so the ΔH_{rxn} must be divided by 2 to get the enthalpy of combustion:

$$\Delta H_{\text{comb}} = -4519\text{ kJ/mol}$$

The ΔH_f for $H_2O(g)$ provided on the data sheet is not used in this calculation. If a student uses that value instead of the ΔH_f for $H_2O(l)$ but otherwise does the problem correctly, they will get –4321 kJ/mol. The only answer close to this is the correct one.

C07. (D) $E = h\nu$, and $\nu\lambda = c$. These equations can be combined to yield $E = hc/\lambda$, which is the energy of one photon of a given wavelength. To get the energy of a mole of photons, the energy of one photon must be multiplied by Avogadro's number. If we are using a value for c that is in meters per second, λ must also be in meters.

$$E = \frac{(6.626 \times 10^{-34}\,\mathrm{J \cdot sec})(3.00 \times 10^{8}\ \mathrm{m/sec})}{588 \times 10^{-9}\ \mathrm{m}} \times 6.022 \times 10^{23}/\mathrm{mol} = 2.04 \times 10^{5}\ \mathrm{J}$$

C08. (A) Lattice energy is defined as the energy given off when oppositely-charged gas phase ions combine to form an ionic bond.

C09. (D) The expression for boiling point elevation is $\Delta T = K_b \cdot m \cdot i$, where K_b is the boiling point elevation constant for water (0.512 °C/m), m is the solute concentration in molality (moles of solute per kilogram of solvent), and i is the van't Hoff factor for the solute, which in this case is 2. Solve for molality using the boiling point elevation expression, then multiply by kg of solvent to get moles of NaCl, then multiply by the molar mass of NaCl to get grams of NaCl.

C10. (A) The temperatures in graphs C and D are going *down* as heat is added, so those are incorrect. The boiling point and melting point in graph E don't match those of ethanol, so graph E does not represent ethanol. Graphs A and B differ in the lengths of the horizontal lines. The lengths of these lines are proportional to the heats of fusion and of vaporization for the compound. For ethanol, ΔH_{vap} (38.56 kJ/mol) is about eight times the value of ΔH_{fus} (4.9 kJ/mol), so the line representing boiling should be about eight times the length of the line representing freezing. This is true in Graph A. The specific heat data is not necessary to solve this problem, but correlates to the slopes of the lines for heating each phase. A lower specific heat corresponds to a steeper slope.

C11. (B) Since $[H_2PO_4^-]$ is 10 times $[HPO_4^-]$, the 10 to 1 ratio can be substituted into the K_a expression:

$$K_{a_2} = \frac{[\mathrm{H^+}][\mathrm{HPO_4^-}]}{[\mathrm{H_2PO_4^-}]} = \frac{[\mathrm{H^+}][1]}{[10]} = 6.2 \times 10^{-8}$$

Therefore $[H^+] = 6.2 \times 10^{-7}$, and the pH is 6.21.

C12. (C) Sodium's outermost electron is in the 3*s* orbital. Therefore n = 3 and $\ell = 0$, so only answer C can be correct. (For $\ell = 0$, m_ℓ always = 0, and the value of m_s could be either +½ or –½ because this electron is unpaired.)

C13. (A) Nickel iodate is $Ni(IO_3)_2$. There are two IO_3^- ions for every Ni^{2+} ion, so the Ni^{2+} concentration is half the IO_3^- concentration: $[Ni^{2+}] = ½(4.55 \times 10^{-2}\ M) = 2.275 \times 10^{-2}$ M.
$K_{sp} = [Ni^{2+}][IO_3^-]^2 = [2.275 \times 10^{-2}][4.55 \times 10^{-2}]^2 = 4.71 \times 10^{-5}$.

C14. (C) First calculate K_C, then calculate K_P from that. Moles of PCl_5 = 1.25 – 0.188 = 1.062. Moles of PCl_3 = moles of Cl_2 = 0.188. The gases are in a 2 L container, so the molar concentrations are 0.531 M for PCl_5 and 0.0940 for PCl_3 and Cl_2.

$$K_C = \frac{[\mathrm{PCl_3}][\mathrm{Cl_2}]}{[\mathrm{PCl_5}]} = \frac{[0.0940][0.0940]}{[0.531]} = 1.664 \times 10^{-2}$$

$$K_P = K_C(RT)^{\Delta n}$$

R = 0.08206 L·atm/mol·K, T = 425 °C + 273 = 698 K. There are two moles of gas products in the balanced equation and only one mole of reactant gas, so Δn = 2 – 1 = 1:

$$K_P = 1.664 \times 10^{-2}(0.08206 \times 698)^1 = 0.953$$

C15. (D)

$$\frac{2000\ \text{Calories}}{1\ \text{day}} \times \frac{1000\ \text{cal}}{1\ \text{Calorie}} \times \frac{4.184\ \text{J}}{1\ \text{cal}} \times \frac{1\ \text{day}}{24\ \text{hr}} \times \frac{1\ \text{hr}}{60\ \text{min}} \times \frac{1\ \text{min}}{60\ \text{sec}} = 96.85\ \frac{\text{J}}{\text{sec}} = 96.85\ \text{watts}$$

C16. (C) Volume dispensed = final volume – initial volume. The final volume is 23.60 mL and the initial volume is 20.96 mL. 23.60 – 20.96 = 2.64 mL.

C17. (E) The balanced equation is $4\ NH_3 + 7\ O_2 \rightarrow 4\ NO_2 + 6\ H_2O$.

C18. (C) Calculate the mass percent lead in PbS, then multiply that by the mass of the sample to determine the mass of lead in the sample.

$$\text{Mass percent Pb in Pbs} = \frac{\text{molar mass of Pb}}{\text{molar mass of PbS}} = \frac{207.20}{207.20 + 32.07} = 0.86597$$

$$0.86597 \times 16.0\ \text{tons} = 13.9\ \text{tons}$$

C19. (B)

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}, \text{so } P_2 = \frac{P_1V_1T_2}{T_1V_2}$$

Most of the answers are given in atm, so convert P_1 from 760 torr to 1 atm before doing the calculation so that the answer comes out in atm. Also convert the temperatures from °C to kelvins. T_1 = 298 K and T_2 = 373 K.

$$P_2 = \frac{P_1V_1T_2}{T_1V_2} = \frac{(1\ \text{atm})(10.0\ \text{L})(373\ \text{K})}{(298\ \text{K})(5.00\ \text{L})} = 2.50\ \text{atm}$$

C20. (A) pH is a log scale, therefore you cannot simply average the two pH values. Calculate total moles of H^+ and total volume, then calculate the resulting $[H^+]$ concentration.

For the pH 2.00 solution, $[H^+] = 10^{-2.00} = 1.0 \times 10^{-2}$ M.
1.0×10^{-2} M × 0.1000 L = 1.0×10^{-3} moles of H^+

For the pH 4.0 solution, $[H^+] = 10^{-4.0} = 1 \times 10^{-4}$ M.
1×10^{-4} M $\times$ 0.1000 L = 1×10^{-5} moles of H^+

Total moles of $H^+ = 1.0 \times 10^{-3} + 1 \times 10^{-5} = 1.01 \times 10^{-3}$ moles
Total volume = 0.1000 L + 0.1000 L = 0.2000 L
Final $[H^+] = 1.01 \times 10^{-3}$ mol /0.2000 L = 5.05×10^{-3} M
Final pH = $-\log(5.05 \times 10^{-3}) = 2.3$

We can keep only one significant digit in the final answer because there is only one significant digit in the least precise measurement, which is the given pH 4.0 value. (In a log value only the digits after the decimal point are significant.)

SELECTED PHYSICS SOLUTIONS – UIL INVITATIONAL B 2018

P01. (A) page 40: "…if big G had been even slightly different in the past, then the energy output of the Sun would have been far more variable than anything the biological, climatological, or geological records indicate."

P02. (B) page 58: "By studying these temperature variations in the CMB [Cosmic Microwave Background] … we can infer what the structure and content of the matter was in the early universe."

P03. (E) page 72: "…where space is curved it can mimic the curvature of an ordinary glass lens and alter the pathways of light that pass through. Indeed, distant quasars and whole galaxies have been 'lensed' by objects that happen to fall along the line of sight to Earth's telescopes."

P04. (A) A white dwarf star is the husk of the core left behind when a low or medium mass star (like the Sun) dies. These smaller stars are unable to fuse heavier elements in their core, and thus only fuse up to the element Carbon, which is the primary element left behind as the white dwarf.

P05. (C) This conversion requires both the distance and the time to be converted. I'll do it all in one step: $32.0\ \frac{mm}{min} * \frac{1.0\ cm}{10.0\ mm} * \frac{1.0\ inch}{2.54\ cm} * \frac{1.0\ foot}{12.0\ inches} * \frac{1.0\ mile}{5280\ feet} * \frac{60.0\ min}{1.0\ hour} = 1.19 \times 10^{-3}\ \frac{miles}{hour}$.

P06. (C) This is a straightforward projectile problem, simplified by the fact that the initial velocity is entirely horizontal. First we need to find the time that the golf ball is flying through the air – this is just the time required for it to reach the ground. Using $y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$, noting the initial height of the ball, and that the velocity in the y-direction starts at zero, we get: $0 = 12.5 + 0 + \frac{1}{2}(-9.8)t^2$. Solving gives: $t^2 = 2.55$ or $t = 1.60\ sec$. Now we can find the horizontal distance travelled in that time. The initial velocity is entirely horizontal, and the horizontal acceleration is zero, so the distance travelled in the x-direction is simply: $x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 = 0 + (3.00)(1.60) + 0 = 4.79\ m$.

P07. (D) Since we are only looking for horizontal acceleration, we need only consider the horizontal component of the applied force: $F_x = Fcos\theta = (20.0)\cos(27) = 17.82\ N$. There is no friction, so this is the net horizontal force. Using Newton's second Law: $F = ma = 17.82 = (12)a$ gives $a = 1.49\ m/s^2$.

P08. (A) First, we need to find the velocity of the clay just before it impacts the dragonfly. It is easiest to do this with energy: $E_{initial} = \frac{1}{2}m_c v_0^2 = E_1 = m_c g h_1 + \frac{1}{2}m_c v_1^2$. This gives: $\frac{1}{2}(0.1)(8)^2 = (0.1)(9.8)(2.7) + \frac{1}{2}(0.1)v_1^2$. Solving for the velocity before impact: $v_1 = 3.33$ m/s. The impact with the dragonfly is perfectly inelastic, since they stick together afterwards. So, using conservation of momentum: $m_c v_1 = (m_c + m_d)v_f$. This gives: $(0.1)(3.33) = (0.1 + 0.03)v_f = (0.13)v_f$, resulting in $v_f = 2.56$ m/s.

P09. (C) Since the coin rolls without slipping, we must consider its rotation as well as its linear velocity. I think this problem is most easily solved with energy, but I will need the height of the starting point. To get the height of the inclined plane requires just a little trigonometry: $h = Lsin\theta = (84.8cm)\sin(45) = 60.0cm$. Now we can use conservation of energy, but we must also include rotational kinetic energy: $E_0 = mgh = E_f = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ Inserting the formula for the rotational inertia of a disk: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\omega^2 = \frac{1}{2}mv^2 + \frac{1}{4}mr^2\omega^2$. Now recall that $v = r\omega$, so we get: $mgh = \frac{1}{2}mv^2 + \frac{1}{4}mv^2 = \frac{3}{4}mv^2$. Cancelling the mass and solving for velocity gives: $v^2 = \frac{4}{3}gh = \frac{4}{3}(9.8)(0.6) = 7.84$, and a final linear velocity of $v = \sqrt{7.84} = 2.80$ m/s.

P10. (D) We can use the initial period of the pendulum to get the length of the pendulum: $T = 2\pi\sqrt{\frac{L}{g}} = 1.75 = 2\pi\sqrt{\frac{L}{9.8}}$, which gives $L = 0.760$ m. In the elevator, the g in the period formula must also include the acceleration of the elevator: $T' = 2\pi\sqrt{\frac{L}{g+a}} = 1.62 = 2\pi\sqrt{\frac{0.76}{9.8+a}}$. It takes a bit of algebra, but solving for acceleration gives $9.8 + a = 11.44$, or $a = 1.64$ m/s^2.

P11. (B) This is relatively simple, if you remember the formula for an adiabatic process: $P_1V_1^{\gamma} = P_2V_2^{\gamma}$. This is not the same as the Boyle's law formula for an isothermal process. Using the adiabatic formula, and noting that we don't even need to convert units: $(3.10)(115)^{1.4} = P_2(285)^{1.4}$. This gives $P_2 = 0.870$ atm.

P12. (E) Kirchhoff's rules are critically important to solving many circuit problems, but writing the equations can often be the hardest part. The loop rule states that the sum of the voltage changes around a closed loop will be zero. A voltage change occurs whenever a circuit element (resistor, battery, etc…) is encountered as you journey around the loop. In this problem, the loop indicated is the one at the bottom of the circuit. There are three elements encountered along the loop: battery V_2, and resistors R_4 and R_5. The path indicated is counterclockwise around the loop, which is the same direction as I_C along the bottom part of the path.

However, I_C is not the current flowing through R_4. We need to use Kirchhoff's node rule at the node beside (to the right of) R_4 to get this current. Let's label the current through R_4 as I_4 and have it travelling to the left (out from the node). Using the node rule then gives $I_C = I_B + I_4$. Therefore $I_4 = I_C - I_B$. Now we can write the loop rule terms: passing from the negative side to the positive side of the battery is a positive step up in voltage, so our first term is $+V_2$. Passing through R_4, we are going in the same direction as I_4, so it is a step down in voltage, so our second term is $-R_4I_4 = -R_4(I_C - I_B)$. Then we get to the last element in the loop: we are travelling in the same direction as I_C, so it is a step down in voltage. Our final term is $-R_5I_C$. These terms must sum to zero, so our Kirchhoff equation is $V_2 - R_4(I_C - I_B) - R_5I_C = 0$.

P13. (A) To solve this problem, we need to calculate the potential difference across which the moving charge travels. The potential is created by the +20.0μC charge, so the potential at $x = -4.00cm$ is given by:
$V_1 = \frac{kQ}{r} = \frac{(8.99 \times 10^9)(20.0 \times 10^{-6})}{(0.03-(-0.04))} = 2.57 \times 10^6\ V.$
And the potential at $x = -2.00cm$ is given by: $V_2 = \frac{kQ}{r} = \frac{(8.99 \times 10^9)(20.0 \times 10^{-6})}{(0.03-(-0.02))} = 3.60 \times 10^6\ V.$
Therefore, the potential difference is $\Delta V = V_2 - V_1 = 1.03 \times 10^6\ V.$
And the energy needed to move the charge is $U = q\Delta V = (6.00 \times 10^{-6})(1.03 \times 10^6) = 6.18\ J.$

P14. (C) The Meissner effect is the exclusion of a magnetic field from a superconductor. Tiny currents set up on the surface of the superconductor and cancel any fields attempting to penetrate the superconducting material.

P15. (A) Analysis of this problem requires Lenz' Law, which states that currents are induced to oppose changes in the magnetic flux. Before the magnet is moved, the north pole is towards the loop – therefore the magnetic flux is initially directed into the page. When the magnet is pulled away, the magnetic flux will decrease. Therefore, by Lenz' Law, the current induced will produce a field to replace that lost flux. In other words, the induced current will create a magnetic field in the same direction that the bar magnet produced before it was pulled away – a field directed into the page. By the right-hand-rule, you can point your right thumb in the direction of the induced field (into the page), and your fingers will curl in the direction of the induced current – clockwise.

P16. (B) The focal length for a concave mirror is positive and equal to half the radius of curvature. In other words, $f = \frac{R}{2} = \frac{60}{2} = 30cm$. The object is at $p = 1.5m = 150cm$. Using the mirror equation, we can get the location of the image: $\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{1}{150} + \frac{1}{q} = \frac{1}{30}$. This leads to $\frac{1}{q} = \frac{1}{30} - \frac{1}{150} = \frac{4}{150}$, or $q = \frac{150}{4} = 37.5cm$.

P17. (D) You must use energy to solve this problem – so first we need to convert some of the Balmer lines into energy differences. What we really need are the energy differences for the n = 3 → 2 transition and for the n = 5 → 2 transition. For the n = 3 to n = 2 transition: $\Delta E_{3\to2} = \frac{1240\ eVnm}{\lambda} = \frac{1240}{656} = 1.89\ eV$. And for the n = 5 to n = 2 transition: $\Delta E_{5\to2} = \frac{1240\ eVnm}{\lambda} = \frac{1240}{434} = 2.86\ eV$. Now, we know that transitions, or combinations of transitions, that start and end at the same energy level undergo the same energy differences, so that $\Delta E_{5\to2} = \Delta E_{5\to3} + \Delta E_{3\to2}$. Thus $2.86 = \Delta E_{5\to3} + 1.89$, giving us $\Delta E_{5\to3} = 2.86 - 1.89 = 0.97\ eV$. Then the wavelength for this transition is $\lambda_{5\to3} = \frac{1240\ eVnm}{\Delta E} = \frac{1240}{0.97} = 1280\ nm.$

P18. (E) This particle has three quarks, so that means it is a Baryon (a Meson has a quark and an anti-quark). Strange quarks have a charge of $\frac{-1}{3}$ and up quarks have a charge of $\frac{+2}{3}$, so two strange quarks and an up quark have a total charge of $\frac{-1}{3}+\frac{-1}{3}+\frac{+2}{3}=0$. All quarks have a spin of $\frac{1}{2}$, but it can be oriented up or down. Clearly the two strange quarks are spin-up and the up quark is spin-down (confused yet?). This means the total spin is $\frac{1}{2}+\frac{1}{2}-\frac{1}{2}=\frac{1}{2}$. Finally, note that there are two strange quarks. A single strange quark carries a strangeness of -1, so two strange quarks will give a total strangeness of -2.

P19. (E) In this experiment, the gravitational force of the mass is balanced by the Hooke's Law force of the spring. Mathematically, $kx = mg$. On the graph, we have the length of the spring, L. This is related to the amount the spring is stretched: $L = x + L_0$ where L_0 is the length of the spring when no mass is hanging from it.
So, $k(L - L_0) = mg$, or $kL = mg + kL_0$, which gives the linear equation: $L = \frac{g}{k}m + L_0$. This is the equation of the line of best fit drawn on the graph. From this we can see that the slope of the line is $slope = \frac{g}{k}$.
The approximate slope of the best fit line on the graph is $\frac{50cm-30cm}{300g-150g} = \frac{20cm}{150g} = \frac{0.2m}{0.15kg} = 1.33\ m/kg$.
Now, the spring constant is $k = \frac{g}{slope} = \frac{9.8\ m/s^2}{1.33\ m/kg} = 7.35\ \frac{kg}{s^2} \approx 7.4\ \frac{N}{m}$.

P20. (C) Since this is a series circuit, the current at one point is the same current throughout the circuit. Thus, the total current in the circuit is 129mA. Since we know the total voltage supplied to the circuit, we can use Ohm's Law to get the equivalent resistance of the circuit: $R = \frac{V}{I} = \frac{9.0\ V}{129\ mA} = 70\ \Omega$. Resistors in series add, so if all of the resistors were functioning properly, then we would expect an equivalent resistance of 110 Ω. Clearly, one of the resistors is bad. If a resistor were open, or if the battery was dead, then no current would flow, which is obviously incorrect. The only choices remaining are that a resistor is shorted.
The only way to sum two of the three resistors and get 70 Ω is with the 50 Ω resistor plus the 20 Ω resistor – so it is likely that those two resistors are functioning properly. Therefore, we may conclude that, most likely, the 40 Ω resistor is shorted.
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